James Gangavarapu

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Education

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science — *2018 – 2020*

Georgia State University — Atlanta, GA
Bachelor of Science in Computer Science and Mathematics minor; Cum Laude — *2013 – 2017*

Experience

Microsoft — Atlanta, GA
Software Engineer I Azure IoT Digital Twins — *Jan 2022 – July 2025*

- Designed and implemented distributed system features, including a distributed semaphore. Extended scale testing frameworks, automated deployment pipelines, and enhanced on-call operations for faster incident detection. Led security modernization, reducing attack surface. Delivered key infrastructure upgrades, mentored new engineers, and recognized for technical depth, autonomy, and a strong growth mindset.

NCR — Atlanta, GA
Software Engineer I — *Sep 2020 – Jan 2022*

- Built next-gen CI/CD micro-services using Go, Pubsub, MongoDB, Bigtable, Docker, Kubernetes, Git Actions on Google Cloud. Built a realtime Notfications service using Dataflow[Java], Bigtable and Mongo Realm Sync.

Georgia Tech — Atlanta, GA
Graduate Teaching Assistant CS 6515 Grad. Algo. and CS 4540 Adv. Algo. — *Jan 2019 - May 2020*

Skills

Languages: Java, Go, C, Python, Typescript, SQL **Cloud**: Azure, GCP **Misc**: Linux, Kubernetes, Dataflow, OpenMPI, Docker, gRPC, Make, Git, Data Structures and Algorithms, Parallel Algorithms, System Design

Projects

* **Credits Based CPU Scheduler[*LEAD ENG*]** Multi processor, work conserving, credits based threads scheduler. Reinvented back off mechanism for dead lock avoidance. Correctness verified using data analysis of execution and run times. One of the best implementation in CS6210 Adv. O.S. https://www.tinyurl.com/y5xwzxl5
* **Flat File System[*LEAD ENG*]** Transactional(commit to disk or undo) append only flat file system using an in memory undo log and an on disk redo log. Write throughput increased by orders of magnitude for random writes. Read throughput unchanged for mmap'ed files. Report: https://tinyurl.com/ybpuwu2s
* **Dynamo Key Value Store[*LEAD ENG*]** Distributed KV store Dynamo [DeCandia et al.] using consistent hashing and gRPC. Features high availability using replication, and eventual consistency using a monotonic versioning scheme(last write wins) inspired by Lamport clocks. Report: https://tinyurl.com/yyblwuv7
* **Tiny File System Service[*LEAD ENG*]** Linux system level, synchronous, and asynchronous Tiny File Compression service(with QoS guarentees). Uses Snappy compression, POSIX IPC and Shared Memory. Report: https://tinyurl.com/y3zhhwa4
* **Parallel Quick Sort using MPI on HPC cluster[*LEAD ENG*]** Skewed uneven distribution of distributed array data across processor nodes due to the choice of pivot. Speedup vs efficiency analysis(w.r.t. to processor scaling and workload problem size scaling) and network communication overhead. Report: https://tinyurl.com/yy78gfh8
* **Parallel Jacobi using MPI on HPC cluster[*LEAD ENG*]** Uniform distribution of Matrix data across processor nodes but large communication overheads for parallel Matrix operations. Speedup vs efficiency analysis(w.r.t. to processor scaling and workload problem size scaling) and network communication overhead. Report: https://tinyurl.com/y5q7x9pw
* **Georgia Tech Database EVA** Helped build a query execution engine, enabling users to query unstructured image/video data using declarative SQL like language. Report: https://tinyurl.com/y4oczm5j
* **Featherweight X10 Deadlock freedom proof in CoQ** Prove Featherweight-X10 is a Deadlock free programming language using inductive proofs on the programming language constructs in CoQ, report: https://tinyurl.com/wq9v2qg
* **Buffer overflow attack** Report: https://tinyurl.com/y3n8jxdm
* **Kernel Coin Attack And Hastad's RSA Broadcast Attack** Report: https://tinyurl.com/y3evy68p
* **Supervised, Unsupervised, Randomized Optimization & Markov Decision Processes ML projects** Reports: https://tinyurl.com/yxlltzzd
* **Low-Rate TCP-Targeted DoS Attack** Report: https://tinyurl.com/y36qrqpn Slides: https://tinyurl.com/yxkswx3l